Kiryana Store QR Menu + Order System

کریانہ اسٹور: کیو آر مینو اور آن لائن آرڈر

Developer Plan · React PWA · Firebase · Cloudflare Pages

1. Stack aur architecture

Hissa	Choice	Note
Frontend	**Vite + React + TypeScript**	Ek hi project mein customer aur admin
Styling	Tailwind CSS	Bare buttons, RTL/Urdu aasaan
PWA	`vite-plugin-pwa`	Install optional, offline menu cache
Database	**Firebase Firestore**	Real-time orders
Admin login	Firebase Auth (Email + Password)	Sirf admin ke liye. Customer ka login nahi
Images	Browser mein WebP convert, phir upload	Section 5
Hosting	**Cloudflare Pages** (GitHub se auto deploy)	Free, tez, SSL
QR	`qrcode` library	Admin panel mein generate/download

```
Customer phone --> Cloudflare Pages (React PWA) --> Firestore
Admin phone/PC --> /admin (Firebase Auth) ---------> Firestore
Order --> Firestore mein save + WhatsApp (wa.me) khule
```

Ahem usool: Order sab se pehle Firestore mein save hota hai, phir WhatsApp khulta hai. Is se agar customer WhatsApp mein "Send" na dabaye ya WhatsApp na ho, tab bhi order admin panel mein aa jata hai.

2. Folder structure (expand karne ke liye)

```
src/
 config/
   store.ts          # dukan ka naam, WhatsApp number, delivery fee, currency
   strings.ts        # sab bilingual text yahan (ur + en)
 lib/
   firebase.ts       # init
   whatsapp.ts       # link builders
   image.ts          # WebP converter
   format.ts         # Rs. formatting
 services/           # SIRF yahi Firebase se baat karein
   products.service.ts
   categories.service.ts
   orders.service.ts
   settings.service.ts
 types/              # Product, Order, Category...
 components/
   ui/               # Button, Bi (bilingual text), BigCard, Toast
   shared/
 features/
   menu/             # category list, product grid, search
   cart/             # cart store, qty selector, cart sheet
   checkout/         # form, success page
   track/            # /order/:id status page
   admin/
     orders/  products/  categories/  settings/  qr/
 routes.tsx
```

Expansion ka raaz: UI kabhi directly Firebase ko call na kare, sirf `services/` ke through. Kal Firebase badalna ho to sirf services badlengi.

3. Bilingual UI (toggle nahi, har line mein dono)

Ek component `<Bi ur="آرڈر کریں" en="Place Order" />`: Urdu bari aur upar, English chhoti aur neeche. Sab text `strings.ts` mein ek jagah rahe:

```
export const S = {
  placeOrder: { ur: "آرڈر کریں", en: "Place Order" },
  total:      { ur: "کل رقم",    en: "Total" },
  outOfStock: { ur: "ختم ہے",    en: "Out of stock" },
};
```

Product ke do naam: `nameUr` aur `nameEn`. Fonts: Noto Naskh Arabic (halka) + system sans-serif, `font-display: swap`. Fonts ko apne server se host karen, Google Fonts par depend na karen.

4. Firestore data model

```
// categories/{id}
{ nameUr, nameEn, imageUrl, sortOrder, active }

// products/{id}
{ nameUr, nameEn, categoryId, imageUrl, thumbUrl,
  price,                    // per unit
  unit: "kg" | "litre" | "piece" | "packet" | "dozen",
  step: 0.25 | 1,           // kg ke liye 0.25, baaqi 1
  inStock: true, active: true, sortOrder,
  searchKeywords: ["chawal","rice","چاول"],
  updatedAt }

// orders/{id}   (id = Firestore auto-ID, lamba aur na-guessable)
{ code: "A7K2",            // chota code jo customer/admin bolen
  customer: { name, phone, address, note },
  items: [{ productId, nameUr, nameEn, qty, unit, price, lineTotal }],
  subtotal, deliveryFee, total,
  status: "new" | "preparing" | "out" | "delivered" | "cancelled",
  statusHistory: [{ status, at }],
  createdAt, updatedAt }

// settings/store  (single doc)
{ nameUr, nameEn, whatsapp: "923001234567", isOpen, deliveryFee,
  minOrder, closedMessageUr, closedMessageEn, logoUrl }

// admins/{uid}  { role: "owner" | "staff" }
```

Zaroori: Order ke `items` mein naam aur qeemat ki **copy** save karen (snapshot), taake baad mein qeemat badalne se purane orders na badlein.

5. Images: khud ba khud WebP (admin ko kuch nahi karna)

Admin sirf photo select ya camera se khichta hai. Code khud kaam karta hai:

```
// lib/image.ts
export async function toWebp(file: File, max = 800, quality = 0.8): Promise<Blob> {
  const bmp = await createImageBitmap(file, { imageOrientation: "from-image" });
  const scale = Math.min(1, max / Math.max(bmp.width, bmp.height));
  const c = document.createElement("canvas");
  c.width = Math.round(bmp.width * scale);
  c.height = Math.round(bmp.height * scale);
  c.getContext("2d")!.drawImage(bmp, 0, 0, c.width, c.height);
  return new Promise(res => c.toBlob(b => res(b!), "image/webp", quality));
}
```

- Do sizes banayen: **main (800px)** aur **thumb (200px)**. Menu list mein thumb chale taake page tez khule.
- Target: main 40-80KB, thumb 5-10KB.
- Admin ko upload ke dauran progress bar aur preview dikhayen.
- Ghalat file (jo image nahi) par saaf Urdu/English error.

Storage ka faisla (is par pehle dhyan den): Firebase Storage ab naye projects mein **Blaze (pay-as-you-go) plan** maangta hai. Free quota phir bhi milti hai, lekin card dena parta hai. Do raste hain:

- **Option A (aasaan):** Firebase Storage + Blaze, aur **Budget Alert** laga den.
- **Option B (Cloudflare mein hi):** Cloudflare **R2** bucket, jahan upload ek chhote Pages Function ke zariye ho jo Firebase login token verify kare. Kharcha lagbhag zero, magar setup thora zyada hai.

Dono ke liye `uploadImage(blob, path)` ek hi function rakhen, taake baad mein badalna aasaan ho.

6. Customer flow (kam taleem walon ke liye)

Routes: `/` menu, `/cart`, `/checkout`, `/done/:id`, `/order/:id` (status dekhna)

1. **QR scan** par seedha menu. Upar dukan ka naam aur logo, "Dukan khuli hai / band hai" ka nishan, bari search bar.
2. **Categories** bari tasveeron ke saath (2 columns). Touch karne par products khulen.
3. **Product card:** bari photo, Urdu naam bara, English chhota, qeemat `Rs. 450 / کلو kg`. Neeche `- [qty] +` bare buttons. Kilo wale items mein `+` dabane par 1/4 → 1/2 → 3/4 → 1 aise barhe, aur likha aaye "1/2 / آدھا کلو kg".
4. **Cart bar:** neeche hamesha ek bara sabz button: `3 items · Rs. 1,250 · دیکھیں / View`.
5. **Checkout:** sirf 3 cheezen: **Naam, Phone, Pata** (aur optional "Koi hidayat"). Phone aur naam `localStorage` mein yaad rahen, taake agli baar khud bhar jayen. Neeche bara button "آرڈر کریں / Place Order".
6. **Order bhejna:**
   1. Validation (phone `03XXXXXXXXX` format, naam aur pata khali na hon)
   2. Order Firestore mein save
   3. Cart khali
   4. WhatsApp khulne wala link automatically chale
   5. `/done/:id` page par bara sabz tick: "آپ کا آرڈر مل گیا / Order received", order code, aur do buttons: "WhatsApp par bhejein" (agar auto-open block ho gaya) aur "Order ka status dekhein"
7. **Status page** `/order/:id`: 4 qadam wali patti (Mila → Tayyar → Rasta mein → Pohanch gaya), real-time update.

Ghalti se bachao: Order button do baar dabane par double order na bane (button disable + loading). Cart `localStorage` mein rahe, taake page refresh par na ud jaye.

7. WhatsApp integration

Koi paid API nahi, sirf `wa.me` links:

```
// lib/whatsapp.ts
export function orderToAdminLink(adminNumber: string, o: Order) {
  const lines = [
,`New Order* #${o.code} / نیا آرڈر* 🛒`
    `👤 ${o.customer.name}  📞 ${o.customer.phone}`,
    `📍 ${o.customer.address}`,
    ``,
    ...o.items.map(i => `• ${i.nameUr} / ${i.nameEn} × ${i.qty} ${i.unit} = Rs.${i.lineTotal}`),
    ``,
    `Delivery: Rs.${o.deliveryFee}`,
,`*Total: Rs.${o.total} / کل*`
  ];
  return `https://wa.me/${adminNumber}?text=${encodeURIComponent(lines.join("\n"))}`;
}

// Admin se customer ko status ka paigham (ek tap)
export function statusToCustomerLink(customerPhone: string, o: Order, msg: string) {
  const intl = "92" + customerPhone.replace(/^0/, "");
  return `https://wa.me/${intl}?text=${encodeURIComponent(msg)}`;
}
```

Admin ko notification do tareeqon se milti hai:

1. Customer ke phone se tayyar message admin ke WhatsApp par jata hai (customer "Send" dabaye to).
2. Admin panel mein order **har haal mein** aata hai, Send dabaye ya na dabaye. Isi liye panel hi "asli record" hai.

Admin to customer: Order card par ek button "Customer ko WhatsApp par batayen". Status ke hisab se tayyar Urdu/English paigham khulta hai, admin sirf Send dabata hai.

Hadood jo developer aur malik ko pata honi chahiye:

- `wa.me` customer ke phone se bhejta hai, is liye bagair customer ke Send dabaye message nahi jata. Isi liye panel alert zaroori hai.
- Khud-ba-khud (bina customer ke) WhatsApp bhejna ho to **WhatsApp Business Cloud API** chahiye. Yeh Phase 2 ka option hai. Is mein Meta business verification, template approval aur per-message fees hote hain.

8. Admin panel

Route: `/admin` (Firebase Auth email/password, login ke baad session yaad rahe).

8.1 Orders screen (sab se ahem)

- Real-time list (`onSnapshot`), naya order upar. Tabs: **Naye | Tayyar | Rasta mein | Mukammal | Cancel**.
- **Naya order aane par:** awaaz (alert sound), screen par bara popup, aur browser tab ke title mein `(2) نیا آرڈر`.
- Order card: code, naam, phone (tap par call), pata (tap par Google Maps), items, total.
- **Aik bara button** jo agla status karta hai: `Naya → "Tayyar karein" → "Rasta mein" → "Pohanch gaya"`. Saath chhota "Cancel" button (confirm ke saath).
- Har status badalne par `statusHistory` mein time save, aur "Customer ko WhatsApp" ka button.
- "Print / Parchi" button (80mm thermal printer ke liye saadah print CSS).

- **Note:** Browser mein awaaz tabhi bajti hai jab admin ne ek baar page par touch kiya ho. Isliye admin panel kholte hi bara button "Alerts shuru karein / Start alerts" rakhen.

8.2 Products screen

- Category ke hisab se list, search, bari "+ Naya saman" button.
- Form: Urdu naam, English naam, category, qeemat, unit, photo (camera ya gallery) jo khud WebP ho jaye.
- Har row mein do switch: **"Hai / ختم"** (stock) aur **"Dikhayen / Chhupayen"**.
- **Qeemat ka quick edit:** Ek page par saare items ki qeematein table mein, jahan seedha number badal kar "Save" dabayen.
- **Bulk upload:** CSV (naam, qeemat, unit, category) import, preview dikha kar. Photos baad mein ek ek kar ke.

8.3 Categories, Settings, QR

- Categories: add/edit, tasveer, ordering (upar-neeche ke arrows).
- Settings: dukan khuli/band, WhatsApp number, delivery fee, minimum order, band hone ka paigham.
- **QR page:** menu ka link (apna domain) se QR banaye, PNG/SVG download, aur "Print ke liye A5 poster" ka saadah template (dukan ka naam + QR + "Scan karen aur order karen").

8.4 Reports (chhoti si shuruat)

Aaj ke orders ki tadaad aur kul rakam, aur pichle 7 din. Is ke ilawa abhi kuch nahi.

9. Firestore Security Rules (bohat zaroori)

```
rules_version = '2';
service cloud.firestore {
  match /databases/{db}/documents {
    function isAdmin() {
      return request.auth != null &&
             exists(/databases/$(database)/documents/admins/$(request.auth.uid));
    }

    match /products/{id}   { allow read: if true; allow write: if isAdmin(); }
    match /categories/{id} { allow read: if true; allow write: if isAdmin(); }
    match /settings/{id}   { allow read: if true; allow write: if isAdmin(); }
    match /admins/{id}     { allow read: if request.auth != null && request.auth.uid == id;
                             allow write: if false; }  // console se manually

    match /orders/{id} {
      allow get: if true;            // id lamba/random hai, sirf link wala dekh sakta hai
      allow list: if isAdmin();      // customer sab orders ki list nahi dekh sakta
      allow create: if request.resource.data.status == 'new'
                    && request.resource.data.items.size() > 0
                    && request.resource.data.items.size() <= 100
                    && request.resource.data.customer.name is string
                    && request.resource.data.customer.phone is string;
      allow update, delete: if isAdmin();
    }
  }
}
```

Aur:

- Customer ke browser mein calculate hone wale total par full bharosa na karen. Admin panel mein total dobara compute karke dikhaya jaye aur ajeeb farq par nishan lage. Asli tasdeeq phone call se hoti hai.
- **Firebase App Check (reCAPTCHA v3)** on karen, taake spam bots orders na bhar den.
- Admin ko `admins` collection mein Firebase Console se pehli baar manually add karen.

10. Performance aur offline

- Products aur categories ek hi baar load hon aur **Firestore persistent cache** (`persistentLocalCache`) on ho. Isse reads bhi kam, aur net kamzor hone par menu phir bhi nazar aaye.
- Images par `loading="lazy"`, width/height fixed (layout na hile).
- Product list ko category ke hisab se load karen, saari ek saath nahi.
- Search mein client-side filter (`searchKeywords` + naam), Firebase search ki zaroorat nahi.
- Target: Android low-end par **Lighthouse Performance 85+**, pehla load 3G par 4 second se kam.
- Service worker: app shell aur images cache; Firestore data ka apna cache.
- **Firestore free quota** (50k reads/din) ko dhyan mein rakhen: admin panel ka orders listener `limit(100)` aur sirf pichle kuch din ke orders ke saath chale.

11. Deployment (GitHub → Cloudflare Pages)

1. GitHub repo, branches: `main` (live), `dev` (testing).
2. Cloudflare Pages se repo connect: Build command `npm run build`, output `dist`.
3. Environment variables (`VITE_FIREBASE_*`) Cloudflare dashboard mein.
4. SPA routing ke liye `public/_redirects`: `/* /index.html 200`
5. Firebase Console mein **Authorized domains** mein Cloudflare domain add karen.
6. Custom domain (jaise `dukanname.pk`) Cloudflare par.
7. Har `dev` branch push par Cloudflare khud **preview link** banata hai, usi par test hoga.
8. Firebase ka scheduled backup (Firestore export) ya kam az kam haftawar manual export.

12. Expand karne ke liye pehle se tayyari

Abhi nahi banana, sirf code is tarah likhen ke baad mein aasaani se jure:

Baad ki khwahish	Pehle se kya rakhen
Online payment (JazzCash/Easypaisa/Raast)	Order mein `paymentMethod` aur `paymentStatus` fields
Auto WhatsApp / SMS notification	Notification ek `notify(order)` function ke andar rakhen
Mazeed dukaanein	Har document mein optional `storeId`, aur `config/store.ts`
Discount/coupon	Order mein `discount` field
Customer ka order history	Phone number se orders ki query (index tayyar)
Delivery boy ka access	`admins` mein `role` (owner/staff/delivery)
Stock ginti	Product mein optional `stockQty`

Baad ki khwahish	Pehle se kya rakhen
Web Push notification	Service worker pehle se maujood hai, FCM baad mein add
Dusri zubanein	Sab strings ek hi file mein

Aur: TypeScript strict mode, ESLint + Prettier, har service function par saaf naam, `README.md` mein setup ke qadam.

13. Kaam ka tarteeb (sprint plan)

Sprint	Kaam	Muddat
0	Repo, Vite + Tailwind, Firebase project, Cloudflare Pages deploy, security rules ka pehla version	2-3 din
1	Types, services, bilingual UI components (`Bi` , buttons, cards), admin login	3-4 din
2	Admin: categories + products CRUD, WebP upload, stock/visible switch	5-6 din
3	Customer menu: categories, products, search, qty selector (1/4 kg steps)	5-6 din
4	Cart, checkout, order save, WhatsApp link, done page, status page	4-5 din
5	Admin orders: real-time list, alert sound, status buttons, customer WhatsApp, print	4-5 din
6	Settings, QR generator, bulk price edit, CSV import, reports	4-5 din
7	PWA (install button, offline), performance, App Check	3 din
8	Testing sasti Android phones par, slow 3G, bug fixing	4-5 din
9	Launch: asli data, QR print, malik ki training	2-3 din

Kul: lagbhag **5-7 hafte** ek developer ke liye.

14. Acceptance checklist

- ☐ QR scan se menu 4 second ke andar khule (4G par)
- ☐ Har screen par Urdu aur English dono nazar aayen, koi toggle na ho
- ☐ Kilo wale item mein 1/4, 1/2, 1 kg asani se chunna ho
- ☐ Customer 3 tap/3 fields mein order de sake
- ☐ Order Firestore mein save ho **aur** WhatsApp khule
- ☐ Admin panel mein naya order awaaz ke saath aaye
- ☐ Admin ek tap se status badal sake, customer apna status real-time dekhe
- ☐ Admin koi bhi photo upload kare, wo khud WebP ban kar 100KB se kam ho
- ☐ "Khatam hai" ek switch se ho
- ☐ Customer `orders` ki list nahi dekh sakta (rules test ho chuke hon)
- ☐ Install button optional ho, bagair install ke sab kuch chale
- ☐ Sasti Android phone (2GB RAM) par smooth chale